Laboratory 01

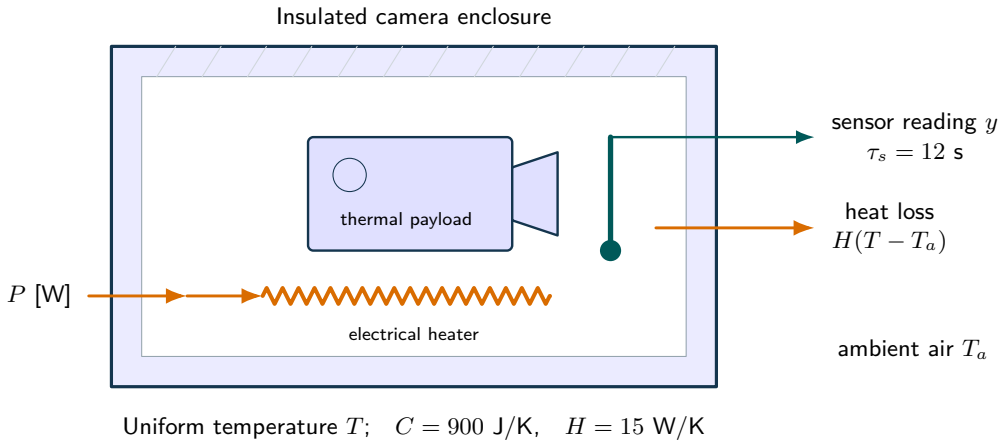
Thermal enclosure and a slow sensor

Worked solutions – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal enclosure: $C = 900$ J/K, $H = 15$ W/K, sensor $\tau_s = 12$ s. Input P is heat rate; x is physical temperature rise and y is the indicated rise. A 40 W step starts from ambient unless a part states another condition.

1. A thermal enclosure under competing design requirements

An insulated instrument box must warm its payload while losing heat through its walls. Use the physical view to identify stored energy, heater input and ambient heat leakage.



Cutaway of a lumped thermal enclosure. Orange arrows carry heat; the probe measures a lagged temperature. The drawing fixes the energy boundary, not a real device geometry.

(a) An enclosure has thermal capacitance $C = 900$ J/K and conductance $H = 15$ W/K. With constant ambient temperature, define $x = T - T_a$ and derive the balance from heat input P to x. Give the units of every term and the zero-state transfer function.

Worked reasoning. Stored incremental energy is Cx. Conservation gives $C\dot{x} = P - Hx$: each term is a heat rate in W because J/s = W. Taking the Laplace transform at $x(0) = 0$ yields $(900s + 15)X = P$, hence $G_x = 1/(900s + 15)$ K/W. The pole is $-H/C = -1/60$ s^{-1} and the time constant is 60 s. The conductance, rather than the capacitance, fixes the static gain.

(b) Keep $C = 900$ J/K but increase H from 15 to 20 W/K. Compare the time constants and the steady rises for 40 W. Find the heater power required to maintain a 3 K rise in each design.

Worked reasoning. Since $\tau = C/H$, the time constant falls from 60 to 45 s. At fixed 40 W, the equilibrium rise falls from 8/3 to 2 K. Maintaining 3 K requires $P = Hx$, giving 45 W and 60 W. Stronger passive heat leakage speeds relaxation and reduces overheating, but raises the steady heater requirement for a prescribed positive temperature rise. These comparisons hold with capacitance and ambient temperature fixed.

(c) A redesigned box must maintain a 3 K rise with at most 45 W and have thermal time constant no greater than 45 s. Keep $C = 900$ J/K and vary only H. Derive both constraints and decide whether any design is feasible.

Worked reasoning. The heater budget requires $3H \leq 45$, or $H \leq 15$ W/K. The time requirement gives $900/H \leq 45$, or $H \geq 20$ W/K. Their intersection is empty. Increasing conductance alone cannot meet both requirements. For example, retaining $H = 15$ would require $C \leq 675$ J/K, whereas retaining $C = 900$ and $H = 20$ would require at least 60 W. These are different redesign choices, not a controller-tuning problem.

2. A reassuring sensor reading can hide stored heat

The nominal box receives a 40 W step from ambient. Its temperature sensor has its own state and a 12 s lag. Evaluate the largest disagreement between the payload and its indicated temperature.

(a) A sensor obeys $12\dot{y} = x - y$, initially $y = 0$. For the 40 W heat step, derive Y/P, $y(t)$ and both initial slopes. Explain the different relative degrees.

Worked reasoning. The sensor transfer is $Y/X = 1/(12s + 1)$, so $Y/P = 1/[15(60s + 1)(12s + 1)]$. Partial fractions give $y = (8/3)[1 - (60e^{-t/60} - 12e^{-t/12})/48]$. The physical state has $\dot{x}(0) = 2/45$, while $\dot{y}(0) = (x(0) - y(0))/12 = 0$. The measured path has relative degree two, so its initial step slope vanishes; the physical path has relative degree one. Both final values are 8/3 K.

(b) For the previous step, find the time and magnitude of the largest positive error $e = x - y$. Verify the error is zero initially and vanishes as time tends to infinity.

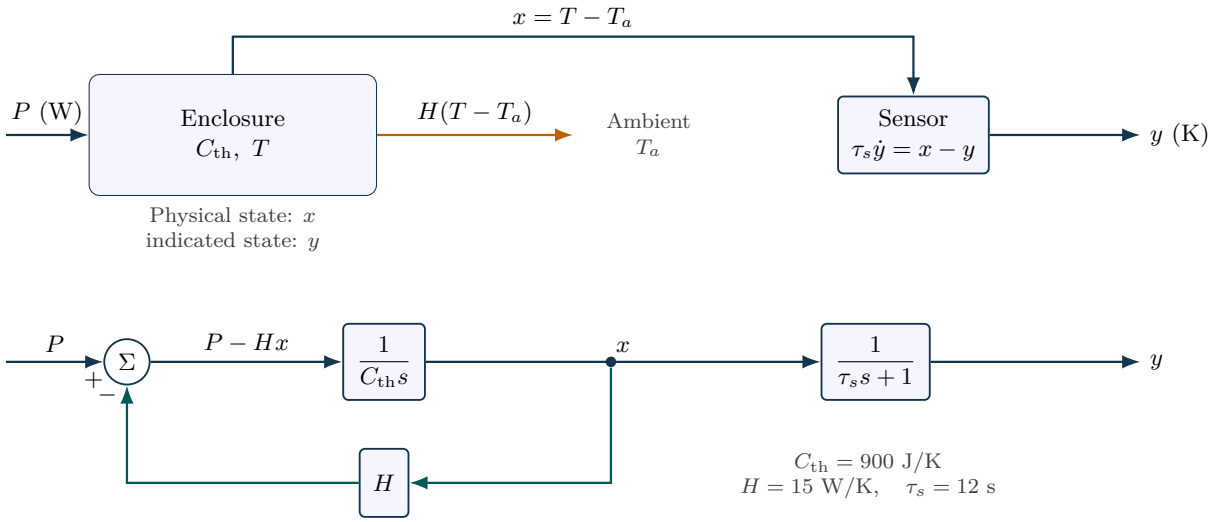
Worked reasoning. Subtract the two responses: $e = (2/3)(e^{-t/60} - e^{-t/12})$. At an interior extremum, $-e^{-t/60}/60 + e^{-t/12}/12 = 0$, giving $e^{-t/15} = 1/5$ and $t_* = 15 \ln 5 \approx 24.14$ s. Substitution gives $e_{\max} = (8/15)5^{-1/4} \approx 0.35666$ K. The error starts and ends at zero; its derivative changes from positive to negative at the unique critical point, proving this is the maximum.

(c) Suppose a limit is checked using only the indicated rise y. Find the smallest constant allowance Δ such that $x(t) \le y(t) + \Delta$ for this particular step. Explain why that allowance is not a universal safety margin.

Worked reasoning. For this stated zero-initial-state step, $x - y$ is nonnegative and its maximum is $(8/15)5^{-1/4} \simeq 0.35666$ K. Thus $\Delta = 0.35666$ K is the smallest uniform allowance for this trajectory. A limit $x_{\max}$ can be checked conservatively by $y + \Delta \le x_{\max}$, but only for the prescribed input and model. Different power histories, initial temperatures, sensor bias or parameter error invalidate a claim that this is a universal bound.

3. A heating pulse with a finite energy budget

The heater can operate at 40 W for only one minute. A 2 K target is desired, followed by passive cooling. Reconcile the continuous-heating prediction with the pulse experiment.



Thermal storage, heat leakage and sensor dynamics are separate elements. The lower loop implements the energy balance, not an added temperature controller.

(a) Starting at ambient, apply $P = 40$ W. Find the equilibrium rise, initial slope and time to reach a rise of 2 K. Is a 3 K target reachable with this constant input?

Worked reasoning. Solve the scalar balance: $x(t) = (40/15)(1 - e^{-t/60})$. Thus $x_\infty = 8/3$ K and $\dot{x}(0) = 40/900 = 2/45$ K/s. Set $2 = (8/3)(1 - e^{-t/60})$, obtaining $e^{-t/60} = 1/4$ and $t = 60 \ln 4 \approx 83.18$ s. A 3 K rise exceeds the equilibrium; the monotone response never reaches it. Waiting longer cannot overcome insufficient steady heating.

(b) Apply 40 W for 60 s and then switch the heater off. Derive the physical temperature rise before and after switch-off, and determine the additional cooling time needed to fall to 0.5 K.

Worked reasoning. For $0 \le t \le 60$, $x = (8/3)(1 - e^{-t/60})$. Continuity gives $x(60) = (8/3)(1 - e^{-1}) \approx 1.68565$ K. For $t \ge 60$, the unforced balance yields $x = x(60)e^{-(t-60)/60}$. Setting this equal to 0.5 gives the additional time $\Delta t = 60 \ln[\frac{16}{3}(1 - e^{-1})] \approx 72.92$ s. The formula describes the physical state; the sensor can continue rising briefly after power is removed.

(c) Compute supplied energy, stored incremental energy and heat lost by the switch-off time. Use the balance to explain why the pulse cannot reach the 2 K target even though continuous heating can.

Worked reasoning. The supplied energy is $E_{\text{in}} = 40(60) = 2400$ J. At switch-off, $x(60) = (8/3)(1 - e^{-1})$, so $E_{\text{stored}} = 900x(60) = 2400(1 - e^{-1}) \simeq 1517.09$ J. Integrating $C\dot{x} = P - Hx$ gives $E_{\text{lost}} = 2400 - E_{\text{stored}} = 2400e^{-1} \simeq 882.91$ J. The peak physical rise is about 1.686 K, below 2 K; the continuous 2 K crossing requires about 83.18 s, after the allowed heating interval.

4. Identifying a thermal process through an imperfect instrument

A synthetic characterization record is available, but the instrument may have unmodeled lag. Assess which parameter conclusions survive that uncertainty and whether inverse filtering fixes the problem.

(a) A deliberately synthetic test gives a 3 K equilibrium rise for a 36 W step. A direct physical-state measurement reaches $1 - e^{-1}$ of its final value after 75 s. Identify H and C. Explain what changes if the dynamic observation comes from an unmodeled slow sensor.

Worked reasoning. The equilibrium relation identifies $H = P/x_\infty = 36/3 = 12$ W/K. For a direct first-order state measurement, $\tau = 75$ s and $C = H\tau = 900$ J/K. A unit-DC-gain sensor preserves the equilibrium inference for H but generally delays the fraction-crossing time. Then 75 s is a property of a cascade, not the enclosure time constant. One must identify the sensor dynamics or fit the joint model before inferring C.

(b) An idealized inverse sensor uses $\hat{x}_k = y_k + 12(y_k - y_{k-1})/\Delta t$. For $\Delta t = 1$ s and independent sensor errors of standard deviation 0.02 K, calculate the standard deviation in $\hat{x}$. Explain the consequence of reducing the sampling interval.

Worked reasoning. The noise contribution is $13n_k - 12n_{k-1}$. Independence makes its variance $\sigma^2(13^2 + 12^2) = 313\sigma^2$, so its standard deviation is $0.02\sqrt{313} \approx 0.35384$ K. In general the multiplier is $\sqrt{(1 + 12/\Delta t)^2 + (12/\Delta t)^2}$. It grows as the sampling interval shrinks. Ideal dynamic inversion can remove lag in a noise-free model while greatly amplifying measurement noise; smoothing or estimation changes that tradeoff.

(c) Design a two-stage identification and validation protocol that distinguishes heat leakage from storage and sensor lag. State which conclusions cannot be obtained from the quoted equilibrium and fraction-crossing observations alone.

Worked reasoning. First use several steady power levels to estimate the slope $1/H$ and check offset or ambient drift. Next characterize the sensor independently, or fit a joint plant–sensor model using sufficiently informative transients. Validate on a different power history with separately generated noise. An equilibrium with a unit-DC-gain sensor can determine H, but a single delayed fraction-crossing time cannot uniquely assign plant and sensor time constants. Direct inverse filtering amplifies noise, so a smooth reconstructed curve is not independent identification evidence.

5. A warm start during an ambient-temperature change

Warm initial conditions and external heating act through different terms of the same balance. Keep the distinction between absolute temperature deviation and rise above the current ambient.

(a) Use absolute temperatures and a 12 s sensor. Derive the transfers from heater power and ambient-temperature deviation to measured temperature. Evaluate the final measured change caused by a 30 W power increment and a simultaneous 2 K ambient increase.

Worked reasoning. The deviation balance is $900\dot{\Delta T} = \Delta P - 15(\Delta T - \Delta T_a)$. Therefore $Y/\Delta P = 1/[15(60s+1)(12s+1)]$ and $Y/\Delta T_a = 1/[(60s+1)(12s+1)]$. The DC gains are $1/15$ and one, respectively. Superposition gives the final measured change $30/15 + 2 = 4$ K. This result assumes the sensor measures absolute enclosure temperature and all initial deviations are zero.

(b) At switch-on the enclosure rise is 1 K, the sensor indicates 0 K, and power becomes 40 W. Derive $x(t)$ and $y(t)$ for the nominal parameters. State why multiplying a transfer function by $40/s$ alone is insufficient.

Worked reasoning. The physical solution is $x = 8/3 - (5/3)e^{-t/60}$. Solving $12\dot{y} + y = x$ with $y(0) = 0$ gives $y = 8/3 - (25/12)e^{-t/60} - (7/12)e^{-t/12}$. Check $x(0) = 1$, $y(0) = 0$, and $\dot{y}(0) = 1/12$ K/s. Transfer functions describe zero-state input response; the nonzero initial thermal energy adds a zero-input response. Both parts must be included to reproduce the stated initial condition.

(c) Combine the two effects: use absolute deviations with $\Delta T(0) = 1$ K, $y(0) = 0$, a 30 W heater increment and a simultaneous 2 K ambient increment. Derive $x(t) = \Delta T(t)$ and the measured deviation $y(t)$, and check initial and final values.

Worked reasoning. The balance is $900\dot{x} = 30 - 15(x - 2) = 60 - 15x$. Hence $x = 4 - 3e^{-t/60}$. Solving $12\dot{y} + y = x$ with $y(0) = 0$ gives $y = 4 - (15/4)e^{-t/60} - (1/4)e^{-t/12}$. Consequently $x(0) = 1$, $y(0) = 0$, $\dot{y}(0) = 1/12$ K/s and both final deviations are 4 K. Here x is an absolute deviation from the original reference, not a rise above the newly shifted ambient.